GHCC Global Health Cost Consortium

Exploring the intervention costs of PMTCT and HTC: explaining and predicting unit costs for implementation scenarios using pooled data from several studies.

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Background

- Sub-Saharan Africa accounts for 71% of the global burden of HIV infection. In these countries, heterosexual sex and vertical transmission are the main modes of HIV transmission[1].
- Program planners are facing an era in which funding levels have begun to wane and even decline.
- Program planners, policy makers and funders need quality cost data to effectively and efficiently implement HIV prevention programs
- This study takes advantage of the collective power of available, published costing studies on PMTCT and HTC interventions to estimate explanatory and predictive models

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Objective

Estimated and project unit costs for each step in the attention cascade, taking into account the level of production across a variety of settings, including countries where no data are available.

Methods

- Econometric analysis of VMMC was the guideline…

Data obtainment

- Searching methods conducted for the GHCC Unit Cost Study Repository
- Papers identified for this topics
- DeCormier et al. (2019) for full details.

Standarization

- Description of variables created in order to gather all the information obtained.
- Multiple imputacion of capital cost
- Identification of influential points
- cost inflation to 2017 USD

Analysis

- Explanatory models and construction of our hypothetical implementations scenarios.
- Levels of data: facility-level and scenario-level
- Predicted model, extrapolations and validations

Types of models estimated

Explanatory models

- Explanation is about understanding relationships
- Adjusted by facility-level variables
- Used to identify relevant determinants
- Hypothetical scenarios

Predictive models

- Understand correlations.
- Correlations are not indications of cause and effect
- Adjusted by scenario-level and country- level variables
- Used to extrapolate unit costs to several implementation scenarios

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Why two types of models?

Level 1: Facilities as a unit of observation

F1, Kenya, rural, hospital, private, **VCT $130**

F2, Kenya, rural, hospital, private, **VCT $250 Unit cost**

F3, Kenya, rural, hospital, public, **PICT $180**

F4, Kenya, rural, hospital, public, **PICT $270**

F5, Kenya, rural, hospital, public, **PICT $150**

Level 2: We combined facilities with the same characteristics to create an scenario

F1 & F2 (scenario 1), Kenya, rural, hospital, private, **VCT $190**

F1 & F2 & F3 (scenario 2), Kenya, rural, hospital, public, **PICT $200**

Methods

- However, PMTCT-HTC are different because…

PMTCT → Test → HIV+ → ARV

HTC → Test → HIV+

Results

Systematic review

203 published and gray literatura studies

43 **(21%)** studies of HTC and PMTCT

31 **(72%)** studies of HTC

6 **(20%)** authors shared their data

518 facilities

12 **(28%)** studies of PMTCT

3 **(25%)** authors shared their data

520 facilities

Inclusion criteria:

1) data were collected from a low- or middle-income country setting
2) Primary (not modeled) data were collected After screening for those criteria
3) Aggregated at the facility level

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Data obtained: Distribution by facility characteristics

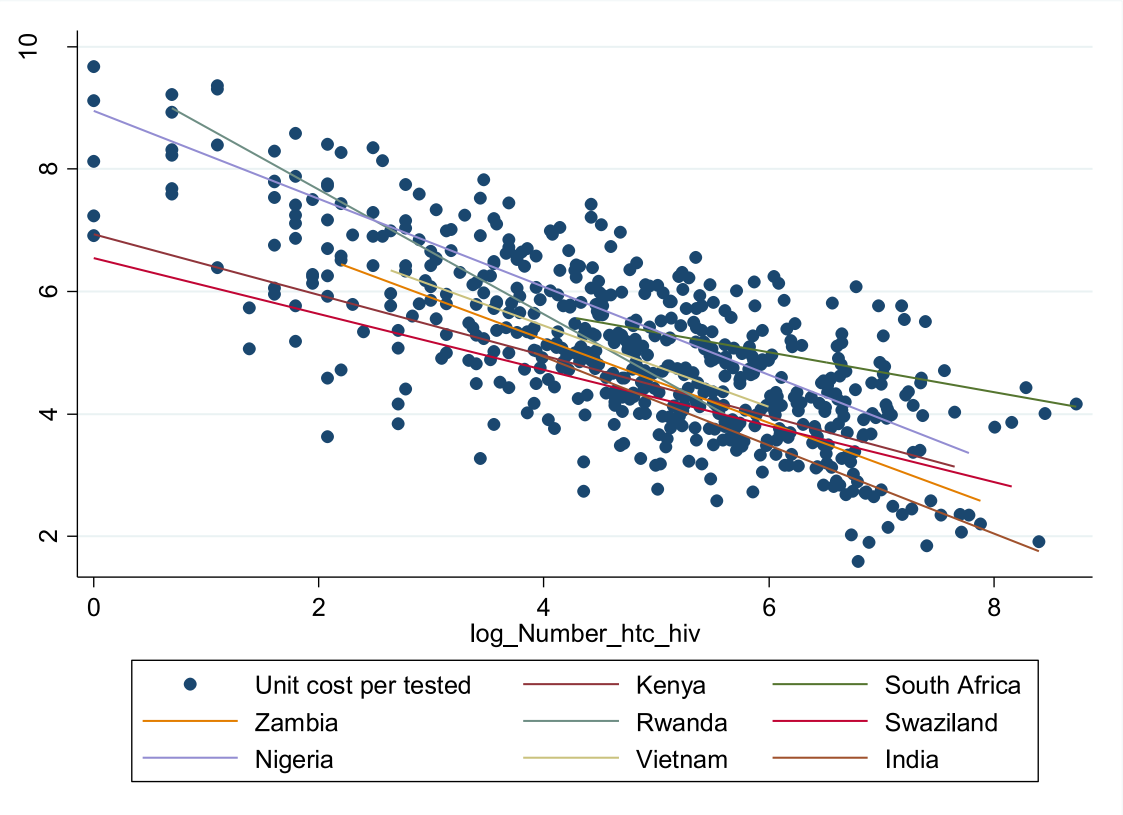
	PMTCT (Primary data)	PMTCT (scenario data)	HTC (Primary data)	HTC (scenarios data)
Observations	Testing: 512	Testing: 78	Testing: 520	Testing: 102
Urbanicity				
Rural Facilities (%)	60	53	51	45
Urban Facilities (%)	40	46	49	55
Ownership				
Private facilities (%)	17	28	18	28
Public facilities (%)	83	72	82	72
Facility type				
Hospitals (%)	41	52	44	42
Clinics (%)	59	47	56	58
Countries	7	7	9	9

Data obtained: Unit costs

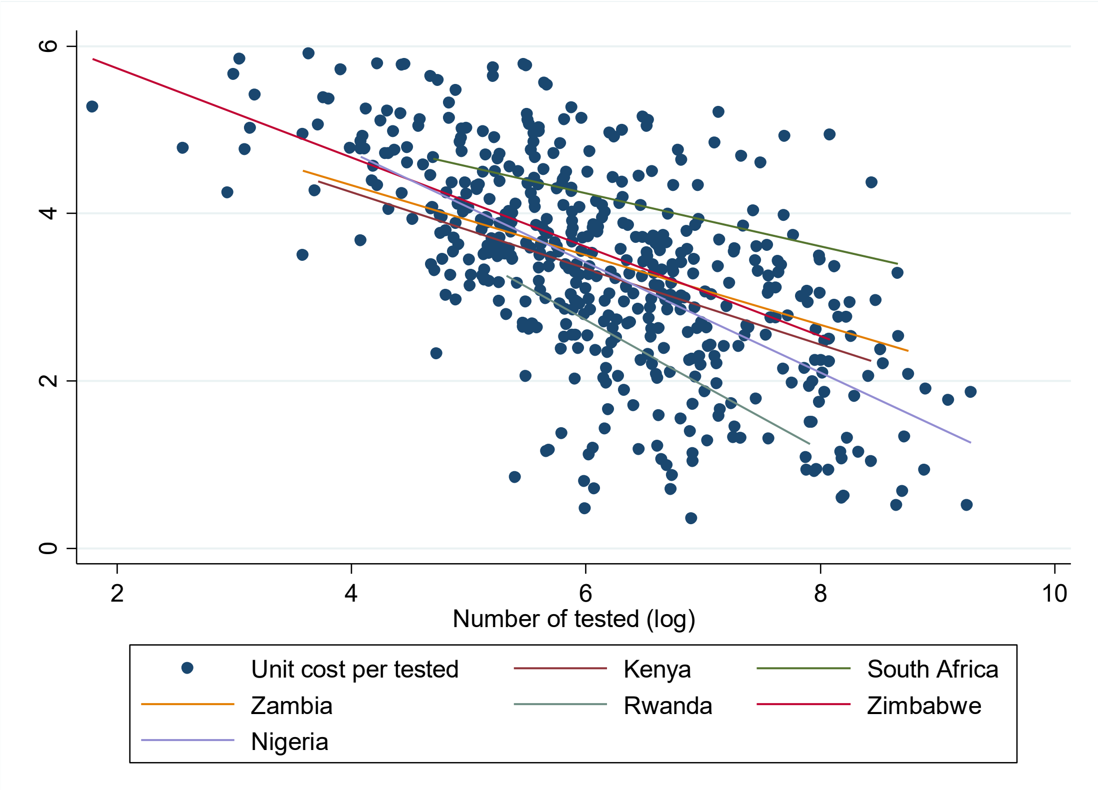
Unit cost	PMTCT (Primary data) Mean (SE)	PMTCT (scenario data) Mean (SE)	HTC (Primary data) Mean (SE)	HTC (scenarios data) Mean (SE)
Unit cost per client tested (USD)	57 (3)	59	17 (.90)	17.9 (1.6)
Unit cost per client HIV+ (USD)	1063 (68)	1129	595 (110)	802 (328)
Unit cost per client HIV+ on ARV (USD)	1418 (87)	1431		
Average number of client tested (USD)	1123	537	3216	2990.6
Average number of client HIV+ (USD)	69	36	366	353.5
Average number of client HIV+ on ARV (USD)	50	24		
Average PMTCT/HTC coverage in 2016 (%)	70	70	60	62
HIV positive rate	0.11 (0.14)		0.13 (0.13)	0.14 (0.10)
Number of studies	3	3	6	6
Countries	7	7	9	9

Unit cost (testing)

HTC



PMTCT



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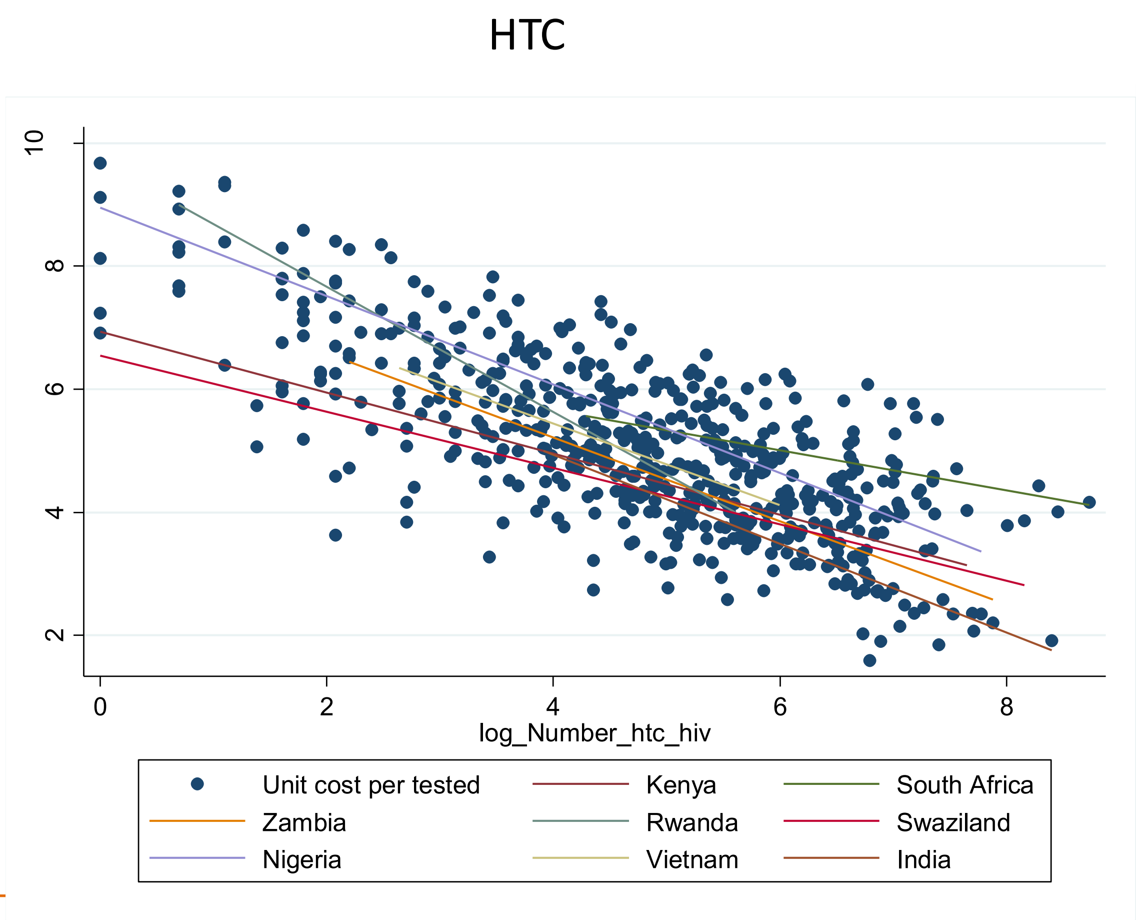
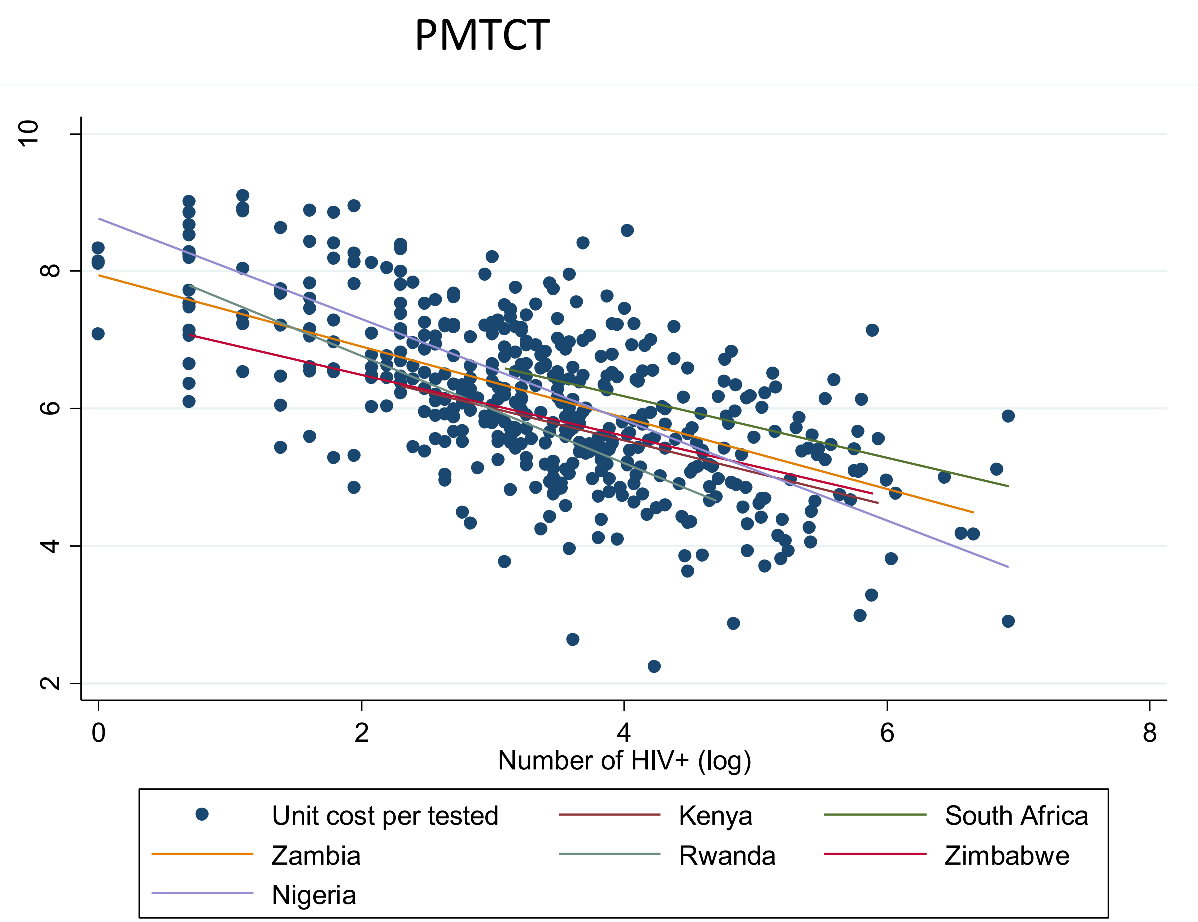
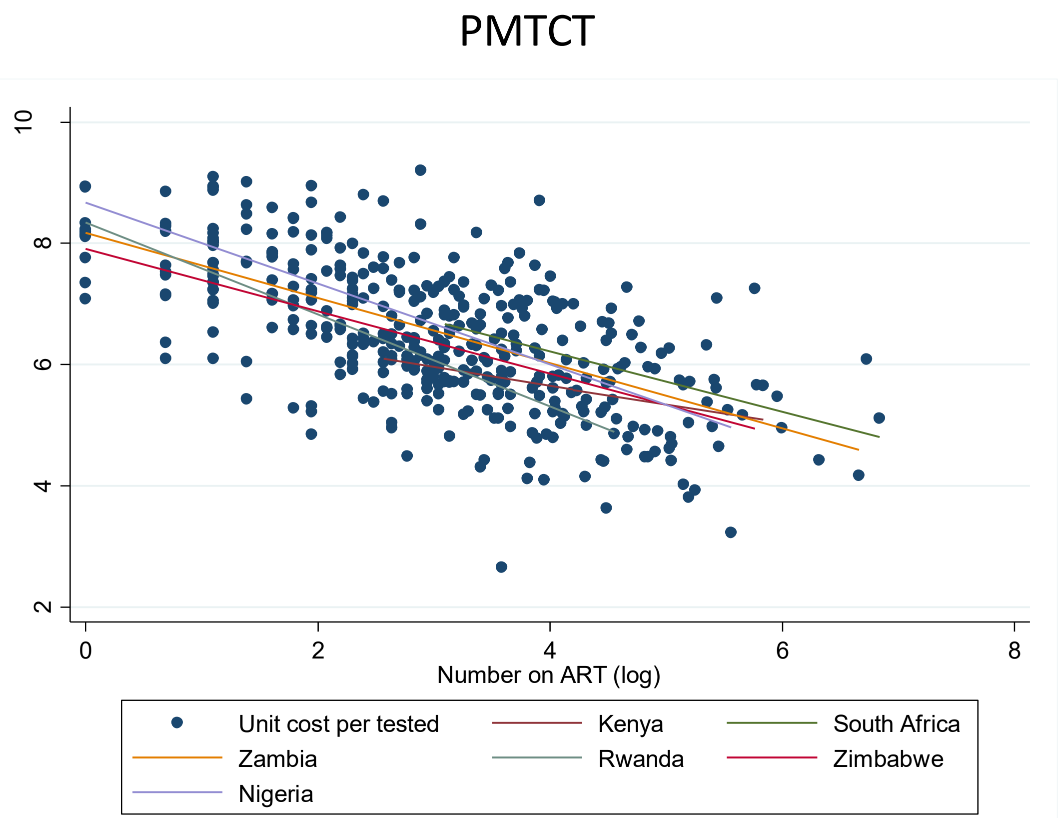
Unit cost (HIV+)

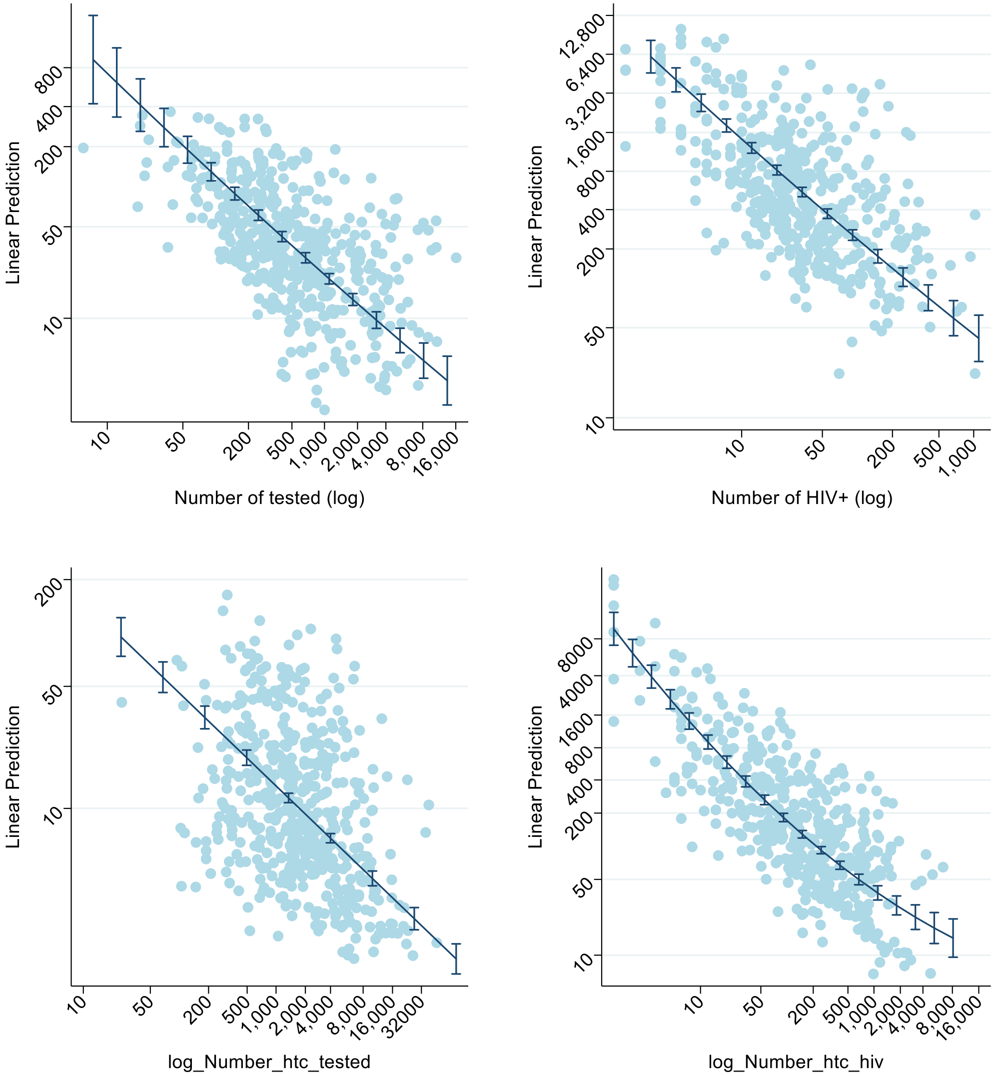




Unit cost (on ARVs)



Unit cost vs Scale



Determinants of unit cost variations (1)

VARIABLES	PMTCT Unit cost tested	PMTCT Unit cost HIV+	PMTCT Unit cost ARV	HTC Test	HTC HIV+
Number of outputs (log)	-0.729***	-0.726***	-0.647***	0.823*	-2.966***
	(0.039)	(0.128)	(0.121)	(0.467)	(0.287)
Number of outputs ^2 (log)	0.014	0.003	-0.007	-0.042***	
	(0.022)	(0.018)	(0.018)	(0.014)	
Public=0; Private=1	-0.589***	-0.504***	-0.425***	-0.209**	-0.217**
	(0.116)	(0.120)	(0.131)	(0.081)	(0.089)
Clinic=0; Hospital=1 = 1, hospital	0.737***	0.572***	0.572***		0.290***
	(0.151)	(0.097)	(0.103)		(0.067)
Urban=0; Rural=1 = 1, Rural	0.287***	-0.085	-0.108	-0.272***	-0.352***
	(0.109)	(0.088)	(0.096)	(0.065)	(0.071)
1.Facility_type#1.Facility_urbanicity	-0.423**	-0.448***	-0.458**	0.275***	
	(0.169)	(0.167)	(0.195)	(0.062)	
Constant	2.402**	3.967***	4.852***	-7.630**	16.368***
	(1.042)	(0.643)	(0.693)	(3.395)	(1.519)
Observations	414	388	337	518	474
R-squared	0.512	0.548	0.55	0.574	0.765

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

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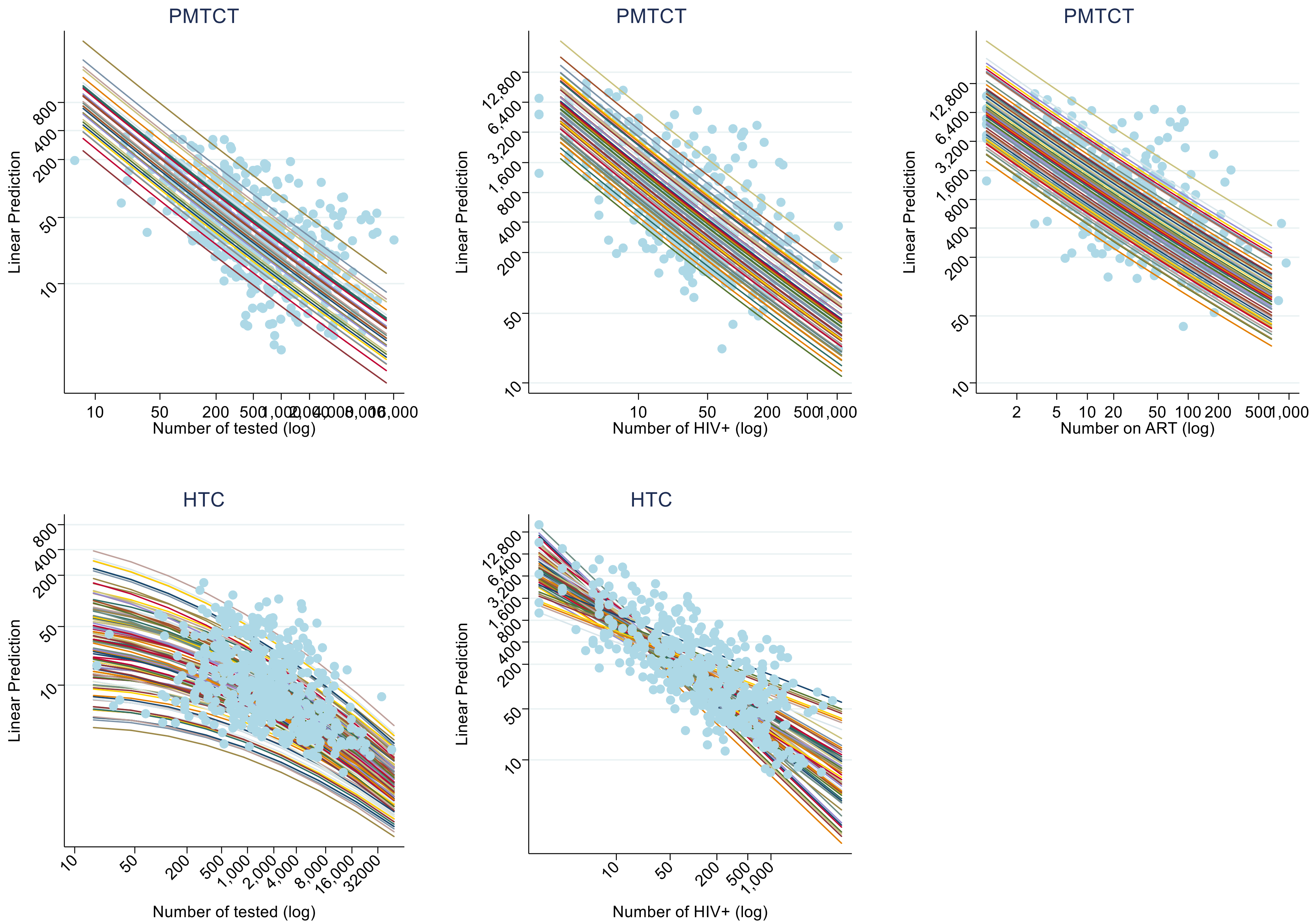
Determinants of unit cost variations (2)

	PMTCT			HTC	
VARIABLES	Unit cost tested	Unit cost HIV+	Unit cost ARV	HTC Test	HTC HIV+
GDP per capita (log)	0.691***	0.523***	0.444***	1.645***	-0.916***
	(0.074)	(0.082)	(0.091)	(0.443)	(0.206)
GDP2017#Number of outputs					
				-0.097*	0.298***
				(0.056)	(0.038)
Year of collection	-0.218***	-0.241***	-0.249***	0.100***	0.108***
	(0.032)	(0.032)	(0.035)	(0.013)	(0.014)
Modality				-0.875***	-0.923***
				(0.096)	(0.123)
Rate of HIV+		-0.226	-0.019		-1.880***
		(0.341)	(0.369)		(0.336)
Retention rate			-0.518***		
			(0.163)		
Constant	2.402**	3.967***	4.852***	-7.630**	16.368***
	(1.042)	(0.643)	(0.693)	(3.395)	(1.519)
Observations	414	388	337	518	474
R-squared	0.512	0.548	0.55	0.574	0.765

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Implementation scenarios


PMTCT
Linear Prediction
Number of tested (log)
10
50
200
400
800
10
50
200
500
1,000
2,000
4,000
8,000
16,000
PMTCT
Linear Prediction
Number of HIV+ (log)
10
50
200
400
800
1,600
3,200
6,400
12,800
10
50
200
500
1,000
PMTCT
Linear Prediction
Number on ART (log)
10
50
200
400
800
1,600
3,200
6,400
12,800
2
5
10
20
50
100
200
500
1,000
HTC
Linear Prediction
Number of tested (log)
10
50
200
400
800
10
50
200
500
1,000
2,000
4,000
8,000
16,000
32000
HTC
Linear Prediction
Number of HIV+ (log)
10
50
200
400
800
1,600
3,200
6,400
12,800
10
50
200
500
1,000

What if there is no information on cost…

Predictive model (1):

VARIABLES	Unit cost PMTCT (tested)	Unit cost PMTCT (HIV+)	Unit cost PMTCT (Linked)	Unit cost HTC (Tested)	Unit cost HTC (HIV+)
Categorical scale of tested = 1, Medium scale of tested	-0.950***	-1.168***	-1.238***	-2.457	-1.475***
	(0.235)	(0.214)	(0.215)	(1.557)	(0.207)
Categorical scale of tested = 2, Large scale of tested	-1.534***	-2.239***	-2.163***	-5.889***	-2.274***
	(0.265)	(0.220)	(0.230)	(1.663)	(0.247)
Clinic=0; Hospital=1	0.218	0.404**	0.482***	0.245**	0.180
	(0.196)	(0.164)	(0.173)	(0.106)	(0.143)
Facility_urbanicity	0.032	-0.258	-0.216	-0.244**	-0.431***
	(0.192)	(0.164)	(0.166)	(0.105)	(0.128)
Public=0; Private=1	-0.040	-0.391**	-0.284	-0.232**	
	(0.216)	(0.176)	(0.183)	(0.115)	
Staff salary index	1.433	-0.965	1.033	0.973**	3.600***
	(1.072)	(0.936)	(1.027)	(0.452)	(0.915)
Constant	0.647	7.588***	9.182***	-1.157	1.328
	(1.270)	(1.237)	(1.320)	(1.237)	(0.917)
Observations	77	74	67	102	88
R-squared	0.376	0.682	0.688	0.704	0.812

Predictive model (2):

VARIABLES	Unit cost PMTCT (tested)	Unit cost PMTCT (HIV+)	Unit cost PMTCT (Linked)	Unit cost HTC (Tested)	Unit cost HTC (HIV+)
(mean) HIV_Prevalence		5.697***	8.076***		2.384**
		(1.626)	(1.698)		(1.105)
(mean) Coverage_pmtct		-0.978**	-1.529***		
		(0.431)	(0.465)		
HTC Coverage					-1.798***
					(0.543)
(mean) log_GDP2017	0.418**	0.071	-0.191	0.434***	0.664***
	(0.164)	(0.159)	(0.168)	(0.163)	(0.137)
1.scale_tested_cat#c.log_GDP2017				0.250	
				(0.205)	
2.scale_tested_cat#c.log_GDP2017				0.602***	
				(0.219)	
Intervention				0.888***	0.938***
				(0.148)	(0.191)
Proportion of HIV+					-4.585***
					(0.657)
Constant	0.647	7.588***	9.182***	-1.157	1.328
	(1.270)	(1.237)	(1.320)	(1.237)	(0.917)
Observations	77	74	67	102	88
R-squared	0.376	0.682	0.688	0.704	0.812

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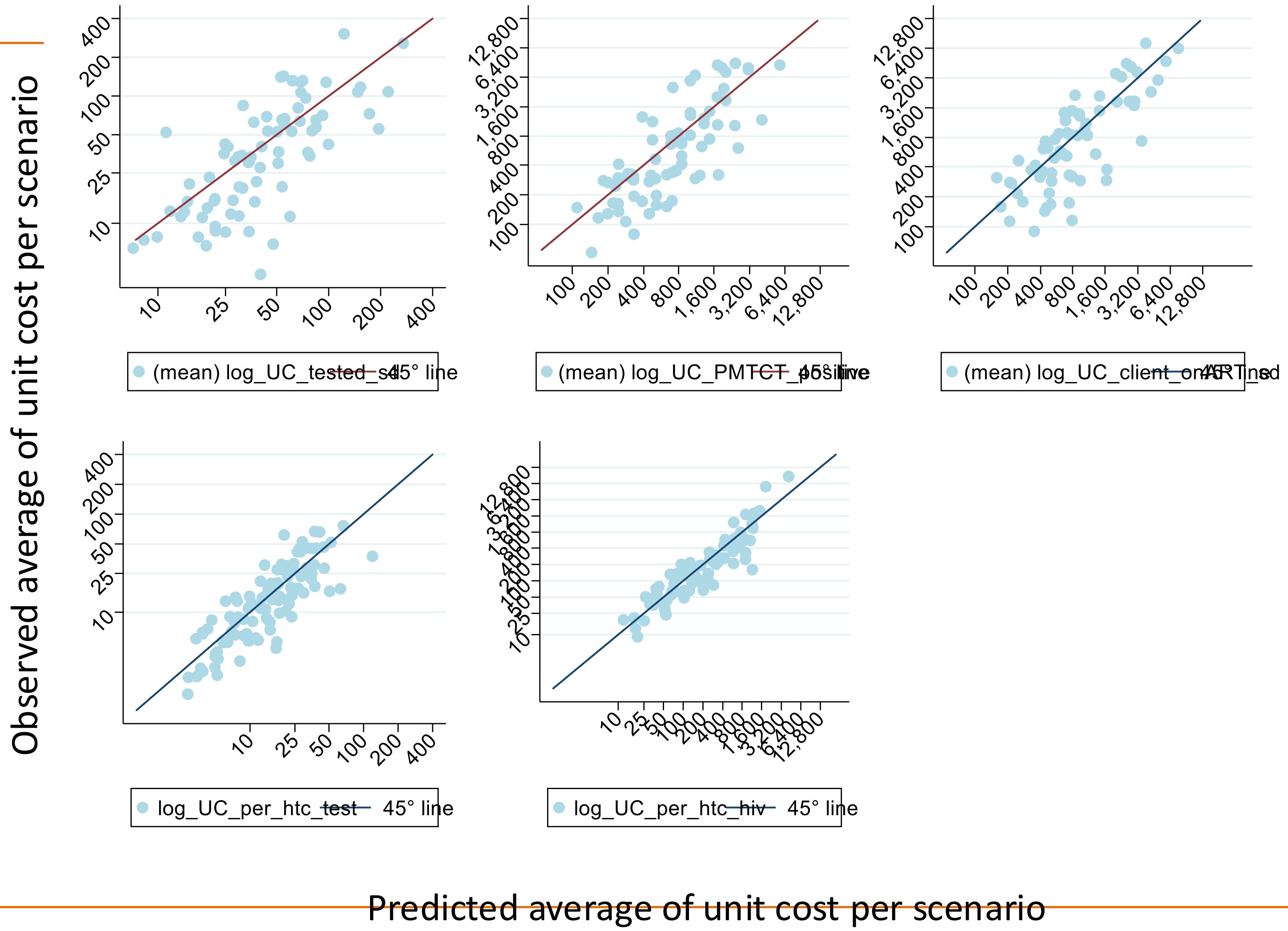
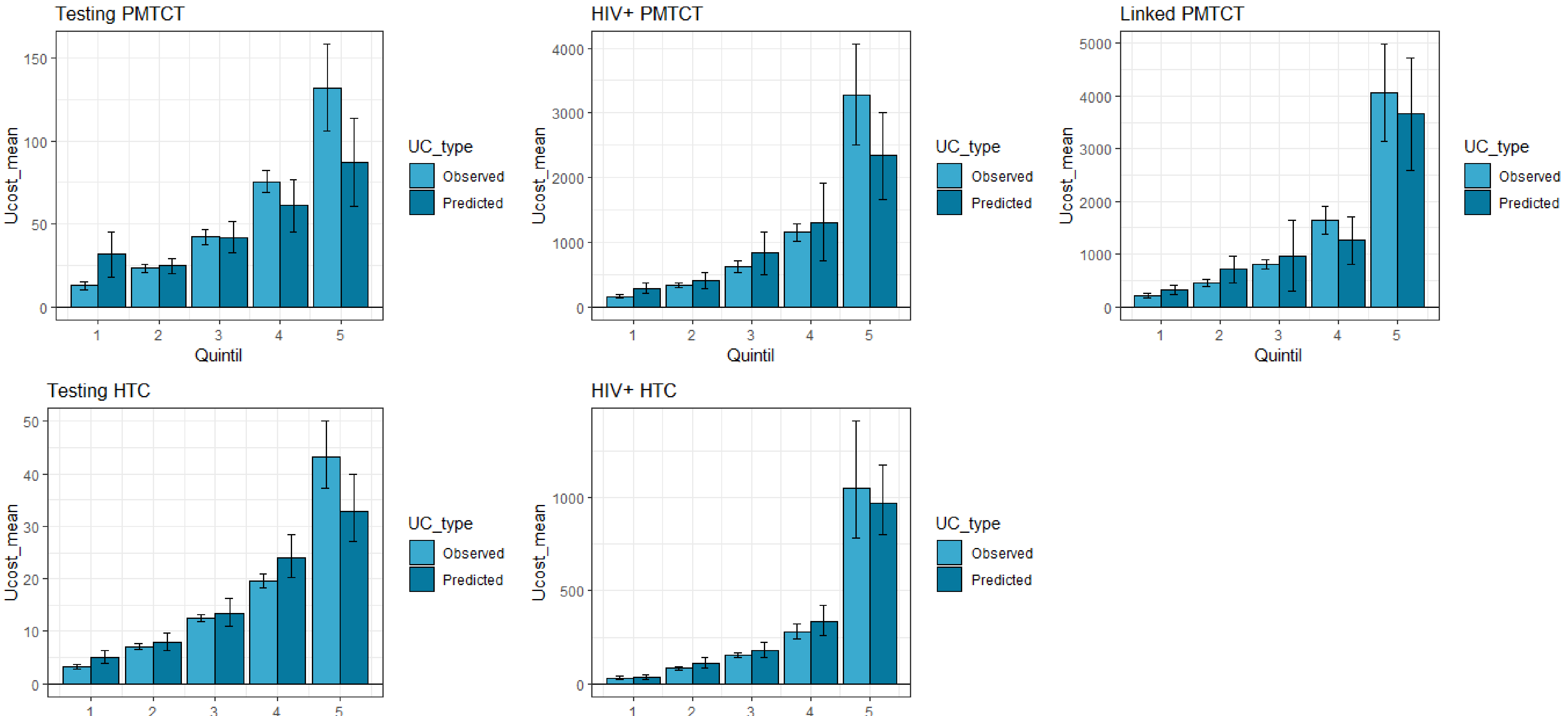
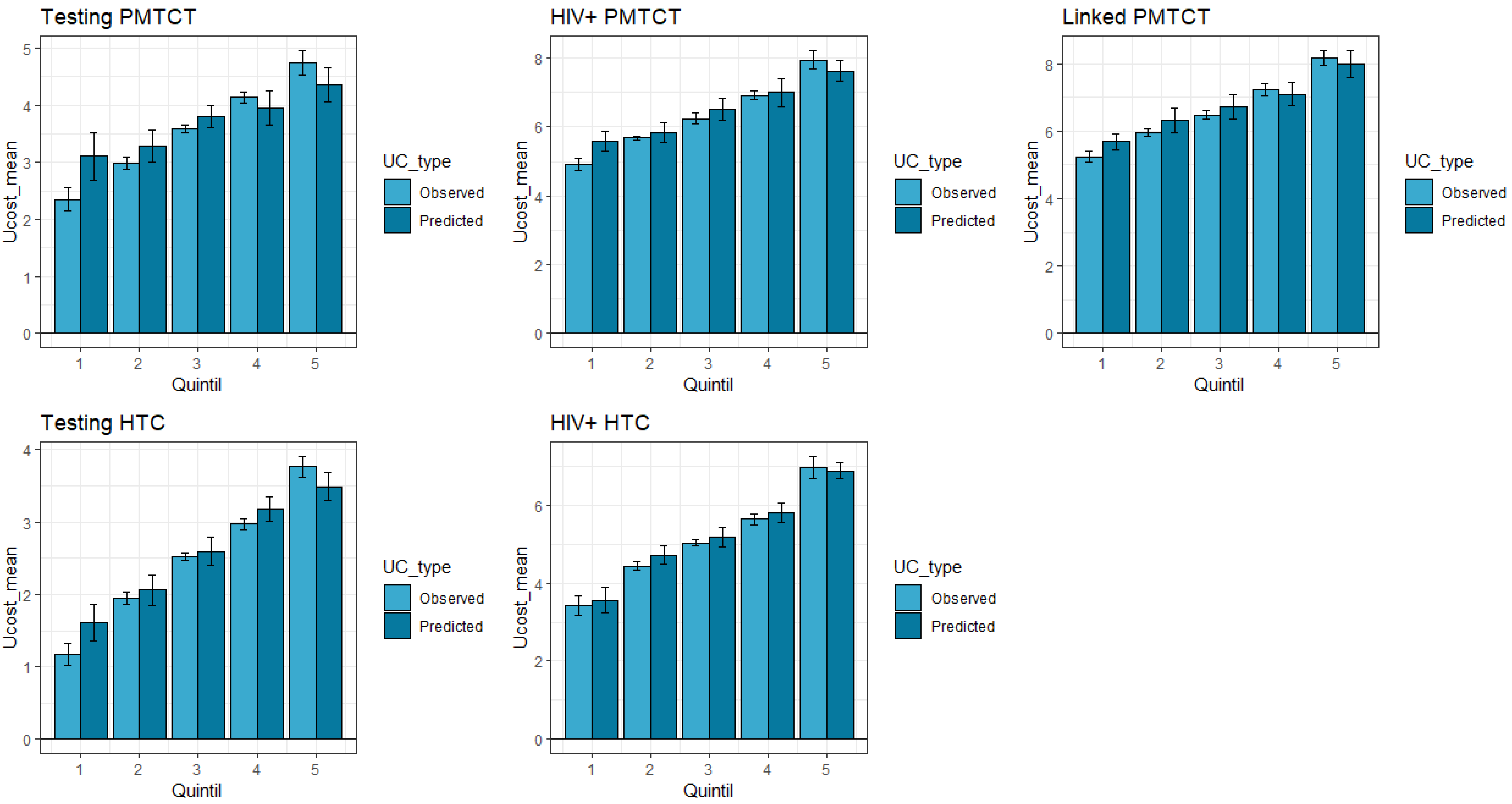
Are our prediction reliable?

Figure 3


Observed average of unit cost per scenario
(mean) log_UC_tested_s
45° line
(mean) log_UC_PMTCT_positive
45° line
(mean) log_UC_client_onART
45° line
log_UC_per_htc_test
45° line
log_UC_per_htc_hiv
45° line
Predicted average of unit cost per scenario

Predictions and validations




Testing PMTCT
HIV+ PMTCT
Linked PMTCT
Testing HTC
HIV+ HTC
Ucost_mean
Quintil
1
2
3
4
5
UC_type
Observed
Predicted

Annex 4 Errors

		mean	min	p10	p25	p50	p75	p90	max	Overlap CI%	Mean Unit cost (USD)
PMTCT TEST	Error in usd	19.91	0.46	1.52	5.68	12.57	29.53	46.21	116.65	75	59
	Error over the mean	34%	1%	3%	10%	21%	50%	78%	198%		
PMTCT HIV+	Error usd	451.73	8.58	30.94	77.69	180.89	700.46	1,562.68	3,009.96	79	1129
	Error over the mean	40%	1%	3%	7%	16%	62%	138%	267%		
PMTCT ART	Error usd	587		48.96	92.43	271.49	850.48	1,400.94		89	1431
	Error over the mean	41%	0%	3%	6%	19%	59%	98%	0%		
HTC TEST	Error usd	6.98	0.06	0.39	1.50	4.24	9.07	17.61	58.94		12
	Error over the mean	0.58	0.01	0.03	0.12	0.35	0.76	1.47	4.91		
HTC HIV+	Error usd	180.85	0.59	5.62	14.53	61.90	129.84	436.73	4829.47		160
	Error over the mean	113%	0%	4%	9%	39%	81%	273%	3018%		

Discussion

- Explanatory models are used to model HTC and PMTCT unit costs for 518 and 521 facilities, respectively
- Predictive models are then employed to extrapolate unit costs for scenarios outside of our sample and we validate these predictions systematically across 9 different countries.
- Our analysis allows us to identify several determinants of unit cost, taking advantage of a large and diverse set of data which includes a variety of contextual characteristics.
- Overall our results show a high level of accuracy in HTC predictions. We mostly attribute this finding to PMTCT as a more-complex program, with a larger original variation of unit cost